

James Dwyer

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PROFESSIONAL SUMMARY

Electrical Engineering student with hands-on experience in PCB design, FPGA-based digital systems, embedded hardware, board bring-up, and laboratory debugging. Designed robot power-distribution and sensor boards in KiCad and Altium, developed Quartus Prime FPGA traffic-control logic using HLSM and VHDL-compatible concepts, and validated STM32, CNC, and motor-driver systems with oscilloscopes, multimeters, structured testing, and power analysis. Applied Python and embedded C++ to debug robotics control algorithms and sensor integration, with additional experience in TCL, Perl, CAD, and manufacturing-process validation.

PROFESSIONAL EXPERIENCE

Union College Robotics Crew

Electrical Engineering Team Lead

Oct 2023 - Present

Schenectady, NY

- Designed custom PCBs in KiCad and Altium, leading cross-functional design reviews for power distribution, motor drivers, and sensor networks in autonomous robots using gaming-hardware product design principles.
- Collaborated with software and mechanical teams using Python and embedded C++ to debug control algorithms, validate sensors, and advance AI-enabled maze navigation for MIT Micromouse robots.

Free Flow Medical

R&D Manufacturing Engineering Intern

Jun 2026 - Sep 2026

Fremont, CA

- Integrated a three-axis CNC control system with closed-loop stepper drives, power distribution, and relay safety circuitry, enabling structured hardware testing and safety verification for reliable operation.
- Developed and validated CNC manufacturing-process improvements through control-system bring-up, structured characterization, and test automation, improving production efficiency and process consistency.
- Designed and iterated SolidWorks CAD models for custom tooling, fixtures, and manufacturing components, supporting CNC machining, production processes, and repeatable manufacturing tests.

Avisi Technologies

R&D Intern

Mar 2026 - May 2026

Redwood City, CA

- Analyzed aqueous humor flow across varying occlusion conditions, strengthening device validation and translating simulation findings into surgical design requirements and performance decisions.
- Prototyped a fiber optic camera system onto surgical forceps, enabling real-time internal visualization and advancing multidisciplinary hardware validation for procedural applications.
- Executed mechanical simulations on the VisiPlate system, evaluating structural integrity, identifying design risks, and optimizing surgical-device performance against hardware requirements.

Residential Elevator Security System

Embedded Systems Developer

Jan 2025 - Mar 2025

- Built and brought up a residential elevator security prototype on an STM32 Nucleo using C++, integrating keypad authentication, LCD feedback, and motor-driver control for authorized access.
- Architected modular, object-oriented C++ software for input handling, motor actuation, and display feedback, enabling scalable embedded performance and repeatable hardware-software validation.
- Debugged and validated driver circuits with multimeters and oscilloscopes, executing repeatable load tests on lock mechanisms to verify safe operation under real-world conditions.

Robotics Initiative

Robotics Project Lead

Sep 2025 - Nov 2025

- Developed an Arduino-based robotic trash collector in C++ with Arduino IDE, integrating DC motors, motor drivers, and ultrasonic sensors to support autonomous accessibility.
- Configured and tested motor-driver circuitry and power distribution through breadboarding, soldering, and power analysis, improving system stability and extending operational runtime.
- Integrated a Wi-Fi camera module with ESP32 firmware and mobile-app interfaces, validating wireless monitoring and real-time remote-control functionality.

Engineering Team

Custom Traffic Control Processor Designer

Oct 2024 - Nov 2024

- Architected a custom traffic control processor in Quartus Prime using FPGA RTL architecture, integrating datapath, FSM, and controller components within an HLSM for intersection-management simulation.
- Implemented counters, comparators, and logic gates to drive real-time state transitions from vehicle counts and timing thresholds, strengthening digital logic verification and debugging.
- Applied VHDL-compatible hardware description concepts and FPGA-based simulation techniques to validate scalable embedded processor architectures through structured digital system design and hardware simulation.

EDUCATION

Union College

Electrical Engineering; Pre-Med

2027

SKILLS

Skills: KiCad, Altium, SolidWorks, Verilog, MATLAB, Python, C++, JavaScript, Arduino IDE, Quartus Prime, STM32, ESP32, Microcontrollers, FPGAs, PCB Design, Power Distribution, Multimeter Testing, Soldering, CAD Modeling, CNC Machining, CNC Control Systems, Tooling and Fixture Design, Fluid Flow Simulation, Mechanical Simulation, Medical Device R&D, Unigraphics, CATIA, Linux, TCL, Creo, Perl, SoC, design review